Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2023-P2-Model)** **Date: 22-04-2026**

Time: 02.30Pm to 05.30Pm **GTA-26** **Max. Marks: 180**

KEY SHEET

MATHEMATICS

1	**A**	2	**C**	3	**C**	4	**C**	5	**AD**	6	**ABC**
7	**ABD**	8	**5**	9	**5**	10	**8**	11	**85**	12	**8**
13	**6**	14	**16**	15	**1.66 to 1.67**	16	**6**	17	**1.66 to 1.67**		

PHYSICS

18	**A**	19	**B**	20	**C**	21	**C**	22	**AC**	23	**ABD**
24	**BD**	25	**3**	26	**4**	27	**9**	28	**5**	29	**2**
30	**3**	31	**22.22**	32	**2.5**	33	**8**	34	**6**		

CHEMISTRY

35	**A**	36	**A**	37	**D**	38	**C**	39	**ACD**	40	**AC**
41	**BC**	42	**18**	43	**10**	44	**6**	45	**14**	46	**682**
47	**7**	48	**9**	49	**51**	50	**89.00 to 91.00**	51	**296**		

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SOLUTIONS
MATHEMATICS

1. For $-1 <= x < 0$ Given equation reduces to $\left(y-e^{sin^{-1}}x\right)^2=2-x^2$

$y=e^{\sin^{-1}x}\pm\sqrt{2-x^2}$

Let A_1 be the area bounded by given curve and x = - 1

$A_1=\int_{-1}^{0}2\sqrt{2-x^2}\,dx=\frac{\pi}{2}+1$ sq. units

$0<x<1$

$A_2=\int_0^1 2\sqrt{1-x^2}\,dx=\frac{\pi}{2}$ sq. units

2. Given differential equation can be rewritten as

$$x^2\left[y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2\right]-\left[2xy\frac{dy}{dx}-y^2\right]=0$$

$$\Rightarrow\frac{\left[y\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2\right]}{1}-\frac{2xy\frac{dy}{dx}-y^2}{x^2}=0$$

$$\Rightarrow\frac{d}{dx}\left(y\frac{dy}{dx}\right)-\frac{d}{dx}\left(\frac{y^2}{x}\right)=0$$

On Integration, we get $y\frac{dy}{dx}-\frac{y^2}{x}=C_1$

$y^2=v.and\ n\ 2y\frac{dy}{dx}=\frac{dv}{dx}$

$$\frac{1}{2}\frac{dv}{dx}-\frac{v}{x}=C_1\Rightarrow\frac{dv}{dx}-\frac{2}{x}v=2C_1$$

$$\text{I.F.}=e^{-\int\frac{2}{x}dx}=e^{-2\ln x}=\frac{1}{x^2}$$

complete solution

$$\frac{v}{x^2}=\int 2C_1\frac{1}{x^2}dx+C_2$$

$$\frac{y^2}{x^2}=-\frac{2C_1}{x}+C_2\Rightarrow y^2=x^2\left(C_2-\frac{2C_1}{x}\right)$$

$$y^2=x(C_2x-2C_1)$$

$$y=\pm\sqrt{x(C_2x-2C_1)}$$

3. $W_n=B_{n+1}-B_n$

$W_n=[(n+1)^2+(n+1)]-[n^2+n]$

$W_n=[n^2+2n+1+n+1]-[n^2+n]$

$W_n=2n+2$

$$S=\sum_{n=1}^{k-1}(2n+2)=2\left[\frac{(k-1)k}{2}\right]+2(k-1)$$

$S=k^2-k+2k-2=k^2+k-2$

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$k^2+k-2=340 \Rightarrow k^2+k-342=0$

$Solving\ gives\ k=18$

4. SM' and SM' intersect at midpoint of PG (where G is point of intersection of normal at P with major axis)

$G=\left(\frac{8\sqrt{2}}{5},0\right)$

$\left(\frac{41}{10\sqrt{2}},\frac{3}{2\sqrt{2}}\right)$

5. $f(x)\le 0, F'(x) as\ f(x)f(x)\ge cF(x)F'(x)-cF(x)\ge 0$

$e^{-cx}\cdot F'(x)-c\cdot e^{-cx}F(x)\ge 0 \Rightarrow \frac{d}{dx}\left(e^{-cx}\cdot F(x)\right)\ge 0$

$e^{-cx}\cdot F(x)\ge e^{-c(0)}\cdot F(0)\Rightarrow F(x)\ge 0$

$\frac{d}{dx}(e^{-x}g(x))<0 \Rightarrow e^{-x}g(x)$ is a decreasing function

$e^{-x}g(x)<e^{-(0)}g(0)\Rightarrow g(x)<0 \quad (\text{for } x>0)$

$f(x)=g(x) has$ one solution $x=0$

$|x^2+x-6|=f(x)+g(x)$

6. $|\text{adj}(\text{adj}(\text{adj}(\text{adj}A)))|=|A|^{16}=4^8\cdot 5^{16}$

$\Rightarrow |A|=10$

$\Rightarrow x+y+z=10$

$So\ p=36$

As we know that $\int_a^b f(x)dx+\int_c^d f^{-1}(x)dx=bd-ac$,

Where f(x) is increasing and invertible function.

Also $f(b)=d, f(a)=c$

Now,

$g(x)=2+2^{\sin x}, then\ g^{-1}(x)=\sin^{-1}(\log_2(x-2))$

Hence,

$\int_{-\pi/2}^{\pi/2}(2+2^{\sin x})dx+\int_{5/2}^{4}\sin^{-1}(\log_2(x-2))dx=2\pi+\frac{5\pi}{4}$

$\Rightarrow 2\pi+\int_{-\pi/2}^{\pi/2}(2^{\sin x})dx+\int_{5/2}^{4}\sin^{-1}(\log_2(x-2))dx=2\pi+\frac{5\pi}{4}$

$\Rightarrow \int_{-\pi/2}^{\pi/2}(2^{\sin x})dx+\int_{5/2}^{4}\sin^{-1}(\log_2(x-2))dx=\frac{5\pi}{4}$

$Hence\ q=5$

$[\vec{a}\times\vec{b}\quad \vec{b}\times\vec{c}\quad \vec{c}\times\vec{a}]=[\vec{a}\vec{b}\vec{c}]^2=|\vec{c}\cdot(\vec{a}\times\vec{b})|^2=|\vec{c}|^2|\vec{a}\times\vec{b}|^2\cos^2\theta$

The maximum value of $[\vec{a}\vec{b}\vec{c}]^2$ is possible only when $\vec{c}$ *is parallel to* $\vec{a}\times\vec{b}$

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$\Rightarrow \theta = 0 \text{ or } \pi$

Hence, $|(\vec{a}\times\vec{b})\times\vec{c}|^2 = |\vec{c}|^2 |\vec{a}\times\vec{b}|^2 \sin^2\theta = 0$

7. from Column 2

$a^2(0) + 4a \cdot f(0) + 1(1) = 4a + 1$

$4a \cdot f(0) + 1 = 4a + 1 \Rightarrow \mathbf{f(0) = 1}$

From Column 1:

$a^2 \cdot f(-1) + 4a \cdot f(1) + 1 \cdot f(2) = 2a^2 + 16a + 17$

By comparing coefficients of the quadratic in a:

a^2 *coefficient* : $\mathbf{f(-1) = 2}$

a *coefficient* : $4 \cdot f(1) = 16 \Rightarrow \mathbf{f(1) = 4}$

Constantterm : $\mathbf{f(2) = 17}$

$f(x) = Ax^3 + Bx^2 + Cx + D$

By using above conditions $f(x) = x^3 + 2x^2 + 1$

8. Given equation is

$\cos^3 3x + \cos^3 5x = (2\cos x \cos 4x)^3 = (\cos 5x + \cos 3x)^3$

$\Rightarrow 3\cos 3x \cos 5x(\cos 5x + \cos 3x) = 0$

$(\because (a+b)^3 - (a^3 + b^3) = 3ab(a+b))$

$\Rightarrow 6\cos 3x \cos 5x \cos x \cos 4x = 0$

$\Rightarrow \cos x = 0 \text{ or } \cos 3x = 0 \text{ or } \cos 4x = 0 \text{ or } \cos 5x = 0$

For $100 < x < 200, \cos x = 0$ *has no solution*

$\cos 3x = 0 \Rightarrow 3x = 450 \Rightarrow x = 150$

$\cos 4x = 0 \Rightarrow 4x = 450, 630 \Rightarrow x = \frac{450}{4}, \frac{630}{4}$

$\cos 5x = 0 \Rightarrow 5x = 630, 710, 890$

$\Rightarrow x = 126, 142, 178$

Sum of all values of x

$= 150 + \frac{450}{4} + \frac{630}{4} + 126 + 142 + 178 = 906$

sum of digits of $k = 15$

9. $\left(\frac{\pi^2}{4} - \pi \sin^{-1}\frac{x}{2}\right) + \pi \sin^{-1}\left(\frac{x}{2}\right) + \frac{\pi^2}{12}(x^2 + 6x + 8) = \frac{\pi^2}{4}(2k - 1)$

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$$\frac{\pi^2}{4}+\frac{\pi^2}{12}(x^2+6x+8)=\frac{\pi^2}{4}(2k-1)$$

$$-1\le\frac{x}{2}\le 1\Rightarrow -2\le x\le 2$$

$$x^2+6x+14=6k$$

$$\Rightarrow 1\le k\le 5$$

10. Conceptual

11. A : coin tossed twice fell head wise

$$\left.\begin{aligned} &B_1: It\ is\ the\ coin\ C_1; \quad P(B_1)=\frac{1}{3}\\ &B_2: It\ \ is\ the\ coin\ C_2; \quad P(B_2)=\frac{1}{3}\\ &B_3: It\ is\ the\ coin C_3; \quad P(B_3)=\frac{1}{3}\end{aligned}\right\}$$

$$P(A/B_1)=\frac{1}{9};P(A/B_2)=\frac{4}{9};P(A/B_3)=\frac{9}{9}(given)$$

$$\therefore P(B_1/A)=\frac{\frac{1}{9}}{\frac{1}{9}+\frac{4}{9}+\frac{9}{9}}=\frac{1}{14}$$

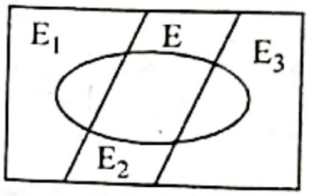
Similarly $P(B_2/A)=\frac{4}{14}$ and $P(B_3/A)=\frac{9}{14}$

Now Let E : event that the same coin tossed again will fall head wise suppose

$$B_1/A=E_1;B_2/A=E_2 and\ B_3/A=E_3$$

$$\therefore P(E)=P(E\cap E_1)+P(E\cap E_2)+P(E\cap E_3)$$

$$=P(E_1)\cdot P(E/E_1)+P(E_2)\cdot P(E/E_2)+P(E_3)\cdot P(E/E_3)$$



$$\therefore \frac{a}{b}=\frac{6}{7}\Rightarrow \quad a=6 and\ b=7$$

$$hence\ a^2+b^2=36+49=85$$

12. $\frac{a_{n+1}}{a_n}-\frac{a_n}{a_{n-1}}=2\left(\frac{1}{n-1}-\frac{1}{n}\right)$

$$\frac{a_{k+1}}{a_k}=3-\frac{2}{k}$$

$$S=\sum_{k=2}^{20}\frac{k}{2}=\frac{1}{2}\sum_{k=2}^{20}k$$

$$S=\frac{1}{2}\left(\frac{20\times 21}{2}-1\right)=\frac{1}{2}(210-1)=\frac{209}{2}=104.5$$

$$[S]=[104.5]=104$$

$$\frac{104}{13}=8$$

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13. $f(x)=(|2x+1|+|x|+|2x+1|)\frac{1}{4}(x^4-1)^2+3$

$Let\ t=\frac{1}{4}(x^2-1)^2+3\ for\ x\in[-\frac{3}{2},\frac{3}{2}], t\in[\frac{3}{4},\frac{73}{64}]$

$t=\frac{1}{4}((x^2-1)^2+3)=1\rightarrow(x^2-1)^2=1$

$\rightarrow x^2-1=\pm1\rightarrow x=\pm\sqrt{2},0$

$\therefore[t]=[\frac{1}{4}(x^2-1)^2+3]$

$$=\begin{cases}0, & x\in(0,\sqrt{2})\cup(-\sqrt{2},0)\\ 1, & x\in[-\frac{3}{2},-\sqrt{2}]\cup\{0\}\cup[\sqrt{2},\frac{3}{2}]\end{cases}$$

$Clearly, f(x) is\ discontinuous\ as\ well\ as\ non-differentiable$

14 & 15

$$=\frac{\cos^2\left(\frac{x}{2}\right)\cdot\cos^2\left(\frac{x}{2^2}\right)\cdot\cos^2\left(\frac{x}{2^3}\right)\ldots\cos^2\left(\frac{x}{2^n}\right)}{\cos x\cdot\cos\left(\frac{x}{2}\right)\cdot\cos\left(\frac{x}{2^2}\right)\ldots\cos\left(\frac{x}{2^{n-1}}\right)}\cdot f\left(\frac{x}{2^n}\right)\pi\left(\frac{8}{\pi}+\frac{8}{\pi}\right)=\pi\left(\frac{16}{\pi}\right)=16$$

$=\frac{\tan x}{x}\cdot f(0)$

$=\frac{2\tan x}{x}$

14. $\pi\left(\frac{8}{\pi}+\frac{8}{\pi}\right)=\pi\left(\frac{16}{\pi}\right)=16$

15. $\frac{(2+2x^2/3)-2(1-x^2/2)}{x^2}=\frac{2+2x^2/3-2+x^2}{x^2}=\frac{5x^2/3}{x^2}=\frac{5}{3}$

16. The equation of line L passing through $A(-7,y_1,z_1)$ is given by

$\frac{x+7}{2}=\frac{y-y_1}{-3}=\frac{z-z_1}{1}$

Comparing this with the provided form

$\frac{x+7}{a}=\frac{y-y_1}{-b}=\frac{z-z_1}{c}$

$a=2,b=3,c=1$

$a+b+c=2+3+1=6$

17. $\frac{x_3-x_2}{2}=\frac{y_3-y_2}{-3}=\frac{z_3-z_2}{1}=\lambda$

$\frac{z_3-z_2}{y_3-y_2}+\frac{x_3-x_2}{z_3-z_2}$

$\frac{1\lambda}{-3\lambda}+\frac{2\lambda}{1\lambda}=-\frac{1}{3}+2=\frac{-1+6}{3}=\frac{5}{3}$

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PHYSICS

18. $N = N_0 e^{-\lambda t}$

$N_2 V = N_1 \Rightarrow V = \dfrac{N_1}{N_2} = e^{\lambda 55}$ ml

19. From the principle of conservation of momentum

$(4u)u_0 + (7u)0 = 10uv + 1uv$

$\Rightarrow v = \dfrac{4v_0}{11}$

loss of $KE = \dfrac{1}{2}(4u)v_0^2 - \dfrac{1}{2}(11u)v^2$

$\Rightarrow 2uv_0^2 - \dfrac{11u}{2} \times \dfrac{16v_0^2}{121} = 2uv_0^2 - \dfrac{8v_0^2}{11}$

$\Rightarrow$ loss of $KE = \dfrac{14uv_0^2}{11}$

$-E\left(1 + \dfrac{m_\alpha}{m_{Li}}\right) = 4.51\text{MeV}$

20. CONCEPTUAL

21. $P = \dfrac{\rho RT}{M}$

22. $V = L\dfrac{di}{dt}$

$\dfrac{dq}{dt} = -n\pi r^2 v_d - i$

$\dfrac{d^2q}{dt^2} = -\dfrac{di}{dt}$

At $t = 0, \quad q = 0 \Rightarrow \phi = 0$

$q = q_0 \sin\left(\dfrac{t}{2\sqrt{\pi\epsilon_0 rL}}\right)$

$i_{max} = 2n\pi r^2 ve$

23. CONCEPTUAL

24. $P = P_{atm} + \dfrac{10}{A_1}$

$P = P_{atm} + \dfrac{20}{A_2}$

25. $f \propto \sqrt{T}$ for strings.

On increasing the tension by 1%

$f' \propto \sqrt{1.01T}$

$\dfrac{f'}{f} = \dfrac{\sqrt{1.01T}}{\sqrt{T}} = (1 + 0.01)^{\frac{1}{2}}$

Beat frequency, $f' - f = f\left(\dfrac{f'}{f} - 1\right) = 1$

Number of beats in 30 seconds = 1 × 30 = 30.

26. When the two spheres come in contact, they exert equal and opposite impulse

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$\int Fadt = I_1(\omega_0 - \omega_1) \quad \ldots(i)$

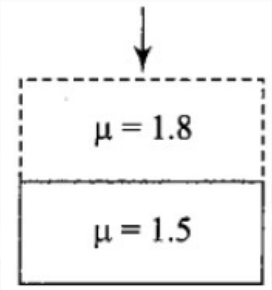
$\int Fbdt = I_2\omega_2 \quad \ldots(ii)$

$Finally \quad \omega_1 a = \omega_2 b \quad \ldots(iii)$

$I_1(\omega_0 - \omega_1) = I_2 \frac{a^2}{b^2}\omega_1$

$\omega_1 = \frac{I_1\omega_0}{I_1 + \frac{a^2}{b^2}I_2} = 4$ rad/s

27. Path difference between rays reflected from upper and lower faces of layer $= 2\mu t \cos r = 2\mu t$ (for normal incidence). But there is change in path of $\frac{\lambda}{2}$ of light at upper surface due to reflection from denser medium. So actual path difference is $2\mu t - \frac{\lambda}{2}$



For constructive interference $2\mu t - \frac{\lambda}{2} = n\lambda$

$t = \frac{(2n+1)\lambda}{4\mu}$ For least thickness n = 0.

$t_{min} = \frac{\lambda}{4\mu} = \frac{648}{4\times 1.8} nm = 90\, nm$

28. $P_1 = P_0\left(1 + \frac{0.2\times 10^{-3}}{V_0}\right)^{5/3}$

$Q = \frac{3}{2}\left[P_0\left(1 + \frac{0.2\times 10^{-3}}{V_0}\right)^{5/3} - P_0\right]V_0$

29. The piston-rod assembly oscillates under the simultaneous actions of gravity, the force applied by atmospheric pressure and the force applied by the pressure of the trapped gas

$$\frac{p_0(A_2 - A_1) + mg}{\sqrt{nmRT}}$$

30. The centre of the wheel is moving with constant speed on a circular path of radius 6R.

Hence, it has a centripetal acceleration of $a_c = \frac{V^2}{6R}$ directed towards the centre of curvature of the convex surface.

With respect to the centre of the wheel, the contact point has acceleration equal to $\frac{V^2}{R}$ directed towards the centre of the wheel.

∴ Acceleration of the contact point in reference frame of ground is

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$$a_P = \frac{V^2}{R} - \frac{V^2}{6R} = \frac{5V^2}{6R}.$$

31. Focal lengths of lenses L_1 *and* L_2 are respectively, given by

$$\frac{1}{f_1} = (\mu_1 - 1)\left(\frac{1}{R} - \frac{1}{\infty}\right) \Rightarrow f_1 = 50 \text{ cm}$$

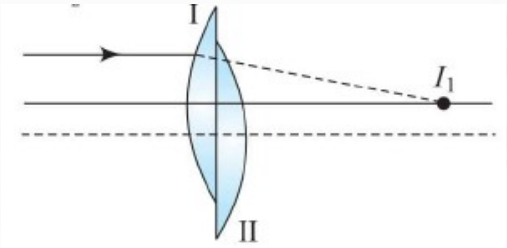
$$\frac{1}{f_2} = (\mu_2 - 1)\left[\frac{1}{\infty} - \left(-\frac{1}{R}\right)\right] \Rightarrow f_2 = 40 \text{ cm}$$

The equivalent focal length f of the combination is determined by

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} \Rightarrow f = \frac{200}{9} \text{ cm}$$

Hence, the image of the parallel beam is formed on the common principal axis at a distance of 22.22 cm from the combination on the right side.

32. Image formed by L_1 is at a distance of 50 cm behind the lens. This image lies on the principal axis of L, and will act as an object for L_2.



For L_2, object distance, u = +50 cm

f_2 = +40 cm

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow v = \frac{200}{9} \text{ cm}$$

Magnification caused by $L_2 m = \frac{v}{u} = \frac{4}{9}$

For L_2 object L_1 is at a distance of 4.5 mm above its principal axis.

Hence, distance of image L_2 of the object (virtual) L_1 } is at a distance (4/9) $\times$4.5 = 2 mm above the principal axis of L_2 [height of image = m $\times$height of object]

Hence, final image is at a distance of 22.22 cm behind the combination at a distance of 2.5 mm below the principal axis of L_1

33. $Mg - T = ma$

$$TR = \frac{MR^2}{2} \cdot \frac{a}{R}$$

From energy conservation

$$mgh = \frac{1}{2}\frac{MR^2\omega^2}{2} + \frac{1}{2}m\omega^2R^2$$

$$\sqrt{\frac{4mgh}{(M+2m)R^2}} = \omega$$

34. $mgh = \frac{1}{2}\frac{MR^2}{2} \cdot \frac{v^2}{R^2} + \frac{1}{2}mv^2$

$$4mgh = Mv^2 + 2mv^2 \Rightarrow v = \sqrt{\frac{4mgh}{M+2m}}$$

which is independent of R

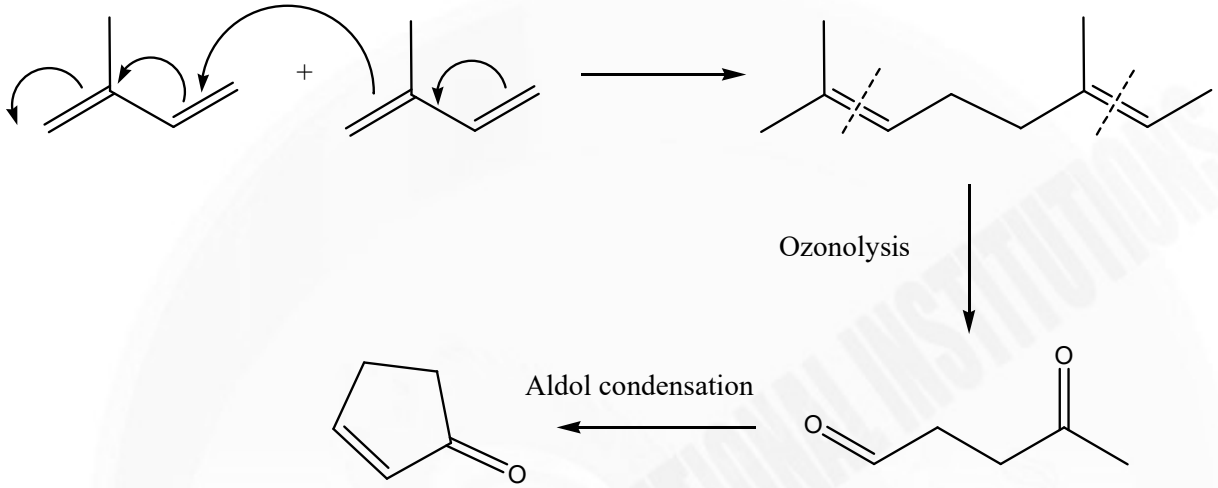
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CHEMISTRY

35. CONCEPTUAL

36. In Tyndall effect, refractive indices of dispersed phase and dispersion medium differ greatly in magnitude. Micelles are formed by surfactant at CMC or above CMC and at Kraft temperature or above Kraft temperature.

37. Key: D



38. Key: C

39. KCl crystallise in rock salt having six nearest neighbor of K^+ *and* Cl^- ion. Packing fraction is 0.74. The number of 2nd nearest neighbor of cation is 12.

40. Key: ABC

41. B and C due to aromatization.

42.
$$(8H^+ + MnO_4^- + 5e^- \rightarrow Mn^{2+} + 4H_2O)\times 2$$
$$(H_2S \rightarrow S + 2H^+ + 2e^-)\times 5$$
$$2MnO_4^- + 16H^+ + 5H_2S \rightarrow 2Mn^{2+} + 5S + 10H^+ + 8H_2O$$
$$2MnO_4^- + 6H^+ + 5H_2S \rightarrow 2Mn^{2+} + 5S + 8H_2O$$
$$x = 8$$
$$y = 10$$

43.

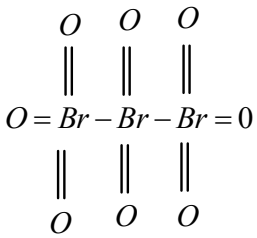
Species	Central Atom	BP	LP	Total ED	Hybridisation	sp3?
$[I_3]^+$	I	2	2	4	sp^3	✓
$[SiO_4]^{4-}$	Si	4	0	4	sp^3	✓
SO_2Cl_2	S	4	0	4	sp^3	✓

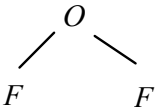
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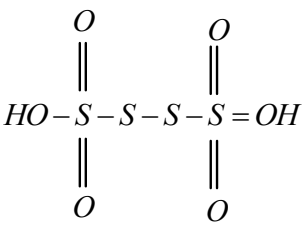
Species	Central Atom	BP	LP	Total ED	Hybridisation	sp3?
XeF_2	Xe	2	3	5	sp^3d	✗
SF_4	S	4	1	5	sp^3d	✗
ClF_3	Cl	3	2	5	sp^3d	✗
$Ni(CO)_4$	Ni	4	0	4	sp^3	✓
XeO_2F_2	Xe	4	1	5	sp^3d	✗
$[PtCl_4]^{2-}$	Pt	4	0	4	dsp^2	✗
XeF_4	Xe	4	2	6	sp^3d^2	✗
$SOCl_2$	S	3	1	4	sp^3	✓
$[ICl_4]^-$	I	4	2	6	sp^3d^2	✗
OSF_2	S	3	1	4	sp^3	✓
$[BH_4]^-$	B	4	0	4	sp^3	✓
IF_5	I	5	1	6	sp^3d^2	✗
$[XeF_5]^-$	Xe	5	2	7	sp^3d^3	✗
$POCl_3$	P	4	0	4	sp^3	✓
$[IBr_2]^-$	I	2	3	5	sp^3d	✗
SF_3^+	S	3	1	4	sp^3	✓
ClO_4^-	Cl	4	0	4	sp^3	✓

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44.

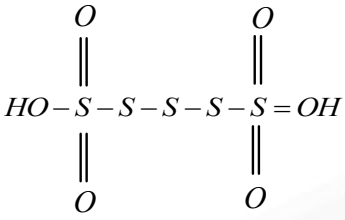
1) 

2) 

3) 

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2 'S' atoms in
zero oxidation states

4) 

3 'S' atoms in
Zero oxidation state

5) $O=C=C=C=O$

1 carbon atom in
zero oxidation states

6) $H_2N=NH_2$

7) H_3PO_3

8) k_2MnO_4

9) Hg_2Cl_2

10) $Fe_2O_3.FeO$

11) $PbO_2.2PbO$

12) $Cl-S-S-Cl$

45. Specific volume $=\dfrac{\text{Volume}}{\text{Mass}}$

$$\text{Mass}=\frac{\pi r^2 H}{\text{Specific Volume}}$$

$$=\frac{3.14\times(75\times10^{-8})^2\times5000\times10^{-8}}{0.75}$$

$$\text{Molar mass}=\text{Mass}\times N$$

$$=7.092\times10^{7}\ \text{g}$$

46.

$$\text{mass of solution}=1.012\times50=50.6\ \text{g}$$

$$0.2=\frac{50.6-x}{60}\times\frac{1000}{x}\qquad x=50\ \text{g}$$

$$\text{wt of urea}=0.6\ \text{g}$$

$$n_{\text{urea}}=\frac{0.6}{60}=0.01$$

$$\text{moles in solution 2}=\frac{0.06}{60}=10^{-3}$$

$$\pi=\frac{(10^{-2}+10^{-3})\times62\times300\times1000}{300}=682\ \text{torr}$$

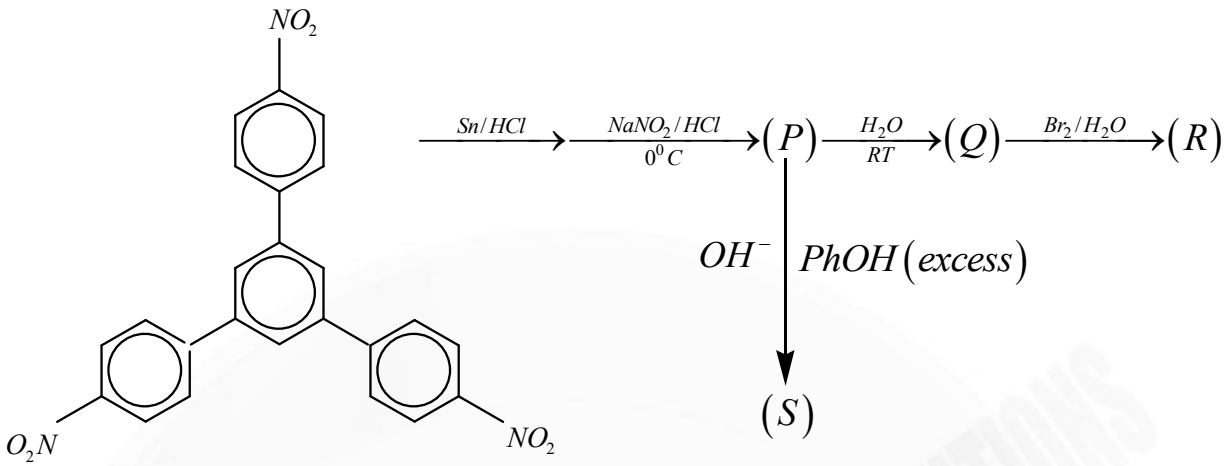
.

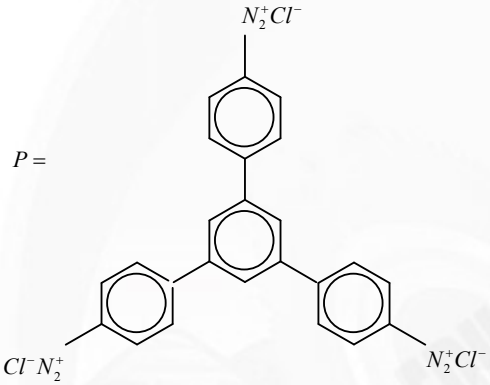
47. 1,3,4,5,6,7,10 are having pH less than phenol.

48 & 49.

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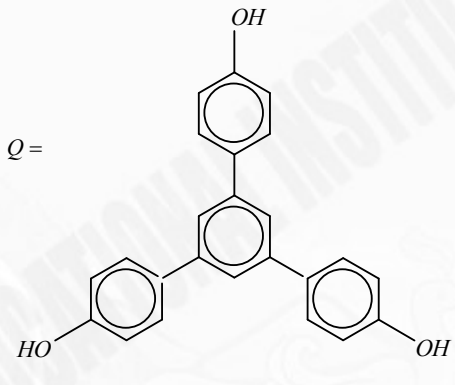
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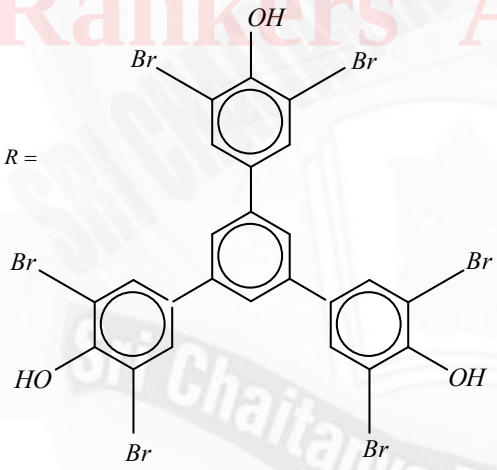
$(C_{24}H_{15}N_6Cl_3)$

493.5 $g\ mol^{-1}$



$(C_{24}H_{18}O_3)$

354 $g\ mol^{-1}$

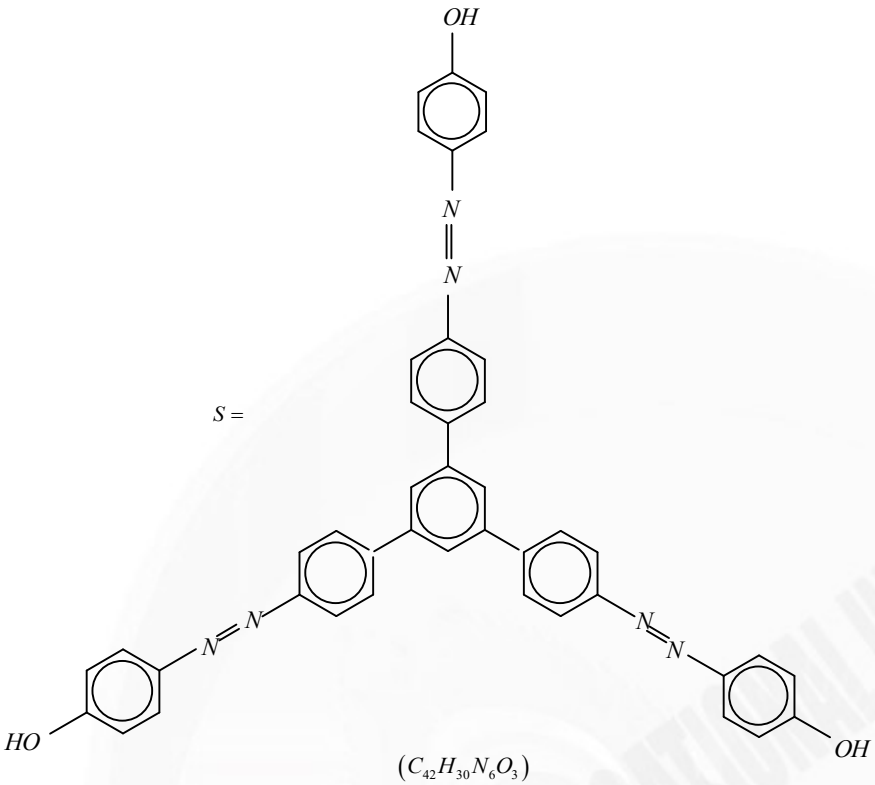


$(C_{24}H_{12}Br_6O_3)$

828 $g\ mol^{-1}$

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$(C_{42}H_{30}N_6O_3)$

$666\ g\ mol^{-1}$

50. $2Hg(g)+O_2(g)\rightarrow 2HgO(s)$

Heat capacity of calorimeter $=20\ kJ\ K^{-1}$

Rise in temperature $=14.8\ K$

$\Delta H^\circ=\Delta U^\circ+\Delta n_gRT$

$=-296-3\times8.3\times298\times10^{-3}$

$\simeq -303.42$ kJ

$\Delta H^\circ=\Delta H_f^\circ(HgO(s))-\Delta H_f^\circ(Hg(g))$

$-303.42=\Delta H_f^\circ(HgO(s))-2\times61.32$

$\Delta H_f^\circ(HgO(s))=-303.42-122.64$

$=-180.78$ kJ

$|\Delta H_f^\circ(HgO(s))|=90.39$ kJ mol^{-1}

51. $\Delta U=z\times\theta=20\times14.8=|296|$ kJ

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